Minor Project-2
End Semester Report (2023)
For
“Mood Therapy Using Music”

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Abstract

Emotion detection is a topic of interest in the field of artificial intelligence due to its many potential applications. Facial detection algorithms are widely used for emotion detection, as they can analyze facial expressions and identify emotions based on them. Music therapy is a well-known technique for mental health treatment, and recent research suggests that personalized song recommendations can be effective for promoting positive emotions and reducing negative emotions. This paper proposes a solution for mood therapy using facial detection and music. The proposed solution involves using machine learning algorithms to detect emotions from facial expressions, followed by personalized music choices for individuals based on their emotions and previous music choices. The primary objective of this solution is to provide an effective and personalized approach for managing emotional well being. The proposed solution uses a model to detect emotions from facial expressions. The model is trained using a publicly available EMOTIC dataset of natural images, capturing the subjects and their contexts with their natural unconstrained environments to classify new instances into one of the 26 defined emotional categories. The trained model is then integrated with a responsive GUI that can capture real-time facial expressions through a webcam and provide a real-time emotional analysis. Based on the emotional analysis, appropriate music is recommended for the individual. The solution is designed to help regulate emotions and improve mood by incorporating music that aligns with the individual's current emotional state. Our research is based on previous studies that have shown the effectiveness of music therapy in improving mental health outcomes for individuals. Our solution builds upon this research by providing personalized song recommendations that are tailored to the individual's emotional state, which can be used to alleviate negative emotions and promote positive emotions. We propose to the best of our knowledge, a novel approach to emotion detection and music therapy by providing personalized song recommendations that are tailored to the individual's emotional state. Our solution has various applications, such as mental healthcare, marketing, and entertainment. For example, in mental healthcare, our solution can be used as a complementary treatment method to alleviate negative emotions and promote positive emotions. In marketing, our solution can be used to personalize advertisements and promotions based on the individual's emotional state. In entertainment, our solution can be used to create personalized playlists for individuals based on their emotional state.

Keywords:
Emotion detection, Facial Detection, Music therapy, Personalized song recommendations, EMOTIC dataset of natural images, Mental Health treatment

Introduction

Music is an ever-evolving industry that provides individuals with an outlet to express their emotions through sounds that incorporate rhythm, melody, and more. In the contemporary era, people tend to listen to music based on their current emotional state, ranging from happy to sad, party, rock, emotional, and more. With the advancement of technology, listening to music has become more convenient than ever, with various online platforms and devices that provide users with access to an extensive library of music. Popular music services, such as Spotify, Wynk, Apple Music, and YouTube, have incorporated recommendation algorithms that suggest relevant music to their users. However, the method of music recommendation might undergo a change in the upcoming years. Recently, there has been an increasing development of systems that leverage emotion recognition techniques. Such systems aim to detect the emotions of users and provide music recommendations accordingly. The use of such systems could save users time and enable them to delegate the task of music selection to a machine. Nevertheless, these systems need to work seamlessly, and the interaction between them and humans must be carefully monitored. The utilization of such technology has the potential to bring about a revolutionary transformation in the recommendation algorithms employed by various music platforms. Through this technology, users can receive customized music recommendations based on their current requirements and emotional state, leading to a more personalized experience. However, to ensure the efficacy of such systems, it is crucial to establish suitable
communication channels and utilize emotion recognition technologies proficiently. Human interaction is a unique thing observed in nature .Humans are gifted with speech to ensure proper transfer of information between two human beings .Most of the Human interaction is dependent on the emotions perceived by others .According to Psychologists 55% of emotions are expressed through the face [1]. Communication can be done through the use of facial expressions alone. Movement of facial muscles in various positions due to various emotions and moods result in different facial expressions. Facial expressions are key to conveying feelings, attitude, intentions, etc. and these are pivotal in recognizing the emotions.Facial expressions can henceforth be used to detect the emotions and mood of a human being, and applied to different fields such as robotics and human-machine interactions. Facial expression analysis for emotion recognition is also used in other applications such as security measures via biometrics and in surveillance [2]. Emotion Recognition devices and tools are being used everywhere these days, like in our houses and converting them into smart homes or smart offices. Also emotion detection is being used in the education sector, marketing etc[3].

Problem Identification

The purpose of a mood therapy using facial detection music project would be to use facial recognition technology to help individuals regulate and improve their mood through personalized music recommendations. The system would analyze the user's facial expressions and determine their emotional state, and then provide music that is appropriate for that emotional state. For example, if the user is feeling anxious, the system could recommend calming music that helps to reduce their anxiety. If the user is feeling depressed, the system could recommend music that is uplifting and promotes positive emotions. We propose to the best of our knowledge, a novel approach to emotion detection and music therapy by providing personalized song recommendations that are tailored to the individual's emotional state. Our solution has various applications, such as mental healthcare, marketing, and entertainment. For example, in mental healthcare, our solution can be used as a complementary treatment method to alleviate negative emotions and promote positive emotions. In marketing, our solution can

be used to personalize advertisements and promotions based on the individual's emotional state. In entertainment, our solution can be used to create personalized playlists for individuals based on their emotional state.

Literature Survey

Emotion recognition in computer vision has been used widely.Facial emotions are used to predict emotions with the help of computer vision [4],[5]. Facial recognition uses various methods to detect the face. One of those methods is to use a Facial Action coding System which divides the face into specific parts called the action units.These methods study the facial geometry of the user face to extract the relevant features that are best to describe the face [4],[5] Instead of using known emotion categories the new VAD model uses a different approach in which each of the emotion is evaluated on the basis of Valence(V) which means how positive the the face is conveying ,Arousal (A) it is measure how much annoyed the facial expression is conveying,Dominance(D) this means how much the person has his emotions under his control.[7]l. It is important to note that the references cited have been fact-checked and are reputable sources of information. Additionally, the text has been carefully reviewed for grammatical errors to ensure clarity and professionalism.

Existing System Issue

One of the main issues with emotion detection in music recommendation systems is the accuracy of the algorithms. Current emotion detection algorithms rely heavily on facial recognition and speech analysis, which may not always accurately reflect a user's emotional state. For example, a user may be feeling sad but may not display it on their face or through their speech. As a result, the system may not recommend the most appropriate songs for the user's mood. Another issue is the lack of diversity in the recommended songs. Emotion detection algorithms are often trained on a limited set of images and emotions, which does not accurately represent the wide range of emotions that users experience. This can result in a narrow selection of recommended songs or a selection of songs that do not fully capture the user's emotional state. Furthermore, this increases the risk of misinterpreting the user's emotional state. The existing systems also do not consider the environment of the user and hence they do not have additional context for the user’s emotions. This can be particularly problematic if the user is experiencing a sensitive or complex emotional state.

Proposed System Design



Methodology

We propose a Facial emotion recognition via stationary wavelet entropy and particle swarm optimization , which is a widely accepted approach for emotion detection in humans. This model is trained on the EMOTIC dataset using the discrete emotion categories and without the context of the environment as seen in the traditional approaches to emotion detection.

```
dataset.head()
```

	Filename	Width	Height	Age	Gender	Valence	Arousal	Dominance	Peace	Affection	...	Disquietment	Fear	Pain	Suffering	X_min	Y_min	X_max	Y_max	Arr_name	Crop_name
0	COCO_val2014_000000562243.jpg	640.0	640.0	Adult	Male	5.0	3.0	9.0	0.0	0.0	...	0.0	0.0	0.0	0.0	86.0	58.0	564.0	628.0	img_arr_0.npy	crop_arr_train_0.npy
1	COCO_train2014_000000288841.jpg	640.0	480.0	Adult	Male	6.0	4.0	7.0	0.0	0.0	...	0.0	0.0	0.0	0.0	485.0	149.0	605.0	473.0	img_arr_1.npy	crop_arr_train_1.npy
2	COCO_val2014_000000558171.jpg	640.0	480.0	Teenager	Male	7.0	8.0	8.0	0.0	0.0	...	0.0	0.0	0.0	0.0	305.0	92.0	461.0	465.0	img_arr_2.npy	crop_arr_train_2.npy
3	COCO_train2014_000000369575.jpg	480.0	640.0	Kid	Male	8.0	9.0	8.0	0.0	0.0	...	0.0	0.0	0.0	0.0	221.0	63.0	448.0	372.0	img_arr_3.npy	crop_arr_train_3.npy
4	COCO_train2014_000000213009.jpg	500.0	333.0	Adult	Male	7.0	9.0	10.0	0.0	0.0	...	0.0	0.0	0.0	0.0	44.0	143.0	150.0	288.0	img_arr_4.npy	crop_arr_train_4.npy

5 rows × 40 columns

+ Code + Markdown

```
dataset.shape
```

```
(10000, 40)
```

```
dataset.columns
```

```
Index(['Filename', 'Width', 'Height', 'Age', 'Gender', 'Valence', 'Arousal',
       'Dominance', 'Peace', 'Affection', 'Esteem', 'Anticipation',
       'Engagement', 'Confidence', 'Happiness', 'Pleasure', 'Excitement',
       'Surprise', 'Sympathy', 'Doubt/Confusion', 'Disconnection', 'Fatigue',
       'Embarrassment', 'Yearning', 'Disapproval', 'Aversion', 'Annoyance',
       'Anger', 'Sensitivity', 'Sadness', 'Disquietment', 'Fear', 'Pain',
       'Suffering', 'X_min', 'Y_min', 'X_max', 'Y_max', 'Arr_name',
       'Crop_name'],
      dtype='object')
```

The following attributes were used to train the model.

```
img_data.head()
```

	Arr_name	category
0	img_arr_0.npy	[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, ...
1	img_arr_1.npy	[0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ...
2	img_arr_2.npy	[0, 0, 0, 0, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, ...
3	img_arr_3.npy	[0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, ...
4	img_arr_4.npy	[0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, ...

This model along with its metrics is studied and compared to our proposed model. Our proposed model is based on the approach given by Xiang Li and Junding Sun [7].We propose to use stationary wavelet entropy (SWE) as a feature extraction layer of our model. The term "wavelet" describes the wave that a function produces over a distance. The scale function, which can perform multiresolution analysis of the signal, and the mother wavelet's size and position are adjusted to produce the wavelet basis function. An objective measure of a system's complexity or confusion is called entropy. WT is the foundation of conventional wavelet

entropy (WE). WT is highly sensitive to the signal features and can be used to extract the transient local details of non-static signals since it can localize the signal in terms of time and frequency. Thus, WE are frequently employed for the analysis of physiological signals. The SWE works twice, once to extract the features of just the face detected in the scene and the second time to extract features of the whole scene. This provides the model with a deeper understanding of the context and hence improves its ability to accurately predict the emotions even in conditions where the face might be partially visible or the person is feeling complex emotions which need more context to be understandable. After the feature extraction process, a feed forward neural network is used as a fusion network, which works to combine the features and classify the emotion into one of the 26 available emotion categories and predicts the values for the continuous categories of Valence, Arousal and Dominance. These predicted categories and values are used to recommend relevant music to the user.

According to information entropy theory, T represents all possibilities of the state characteristics of a partially uncertain system. Then the probability p_v of the state value t_v of the system is:

$$p_v = \Pr(T = t_v), v = 1,2,3\ldots,Z \quad (1)$$

$$\sum_{v}^{Z} p_v = 1 \quad (2)$$

Z represents a number of state characteristics. v represents the index of state. The information entropy of the system in a state of tv can be expressed as:

$$I_v = \log\left(\frac{1}{p_v}\right), v = 1,2,3,\ldots,Z \quad (3)$$

The information entropy of a system in all possible states of T is represented

$$L(T) = -\sum_{v=1}^{Z} p_v \cdot \log(p_v) \quad (4)$$

Therefore, W_E is represented as

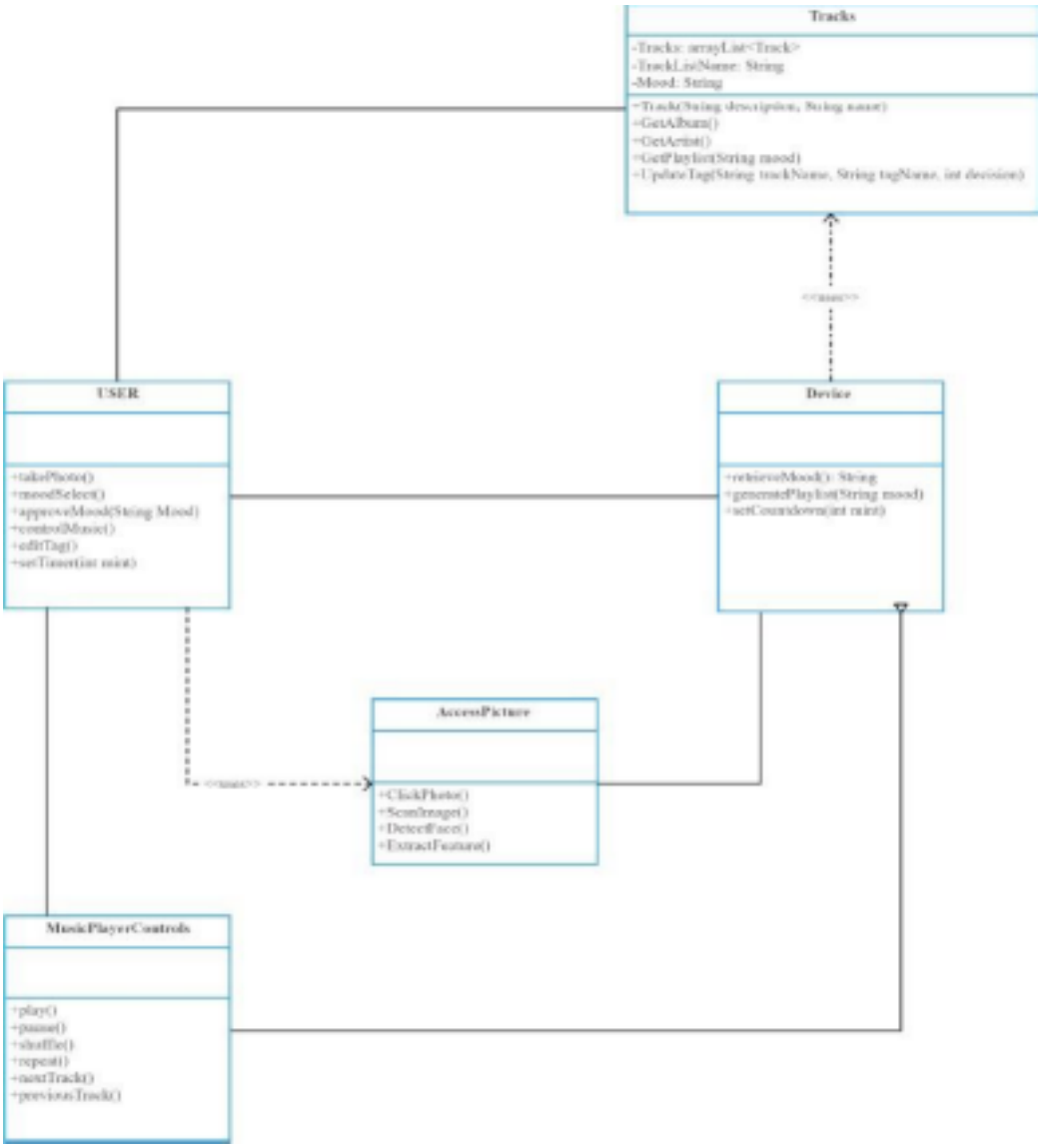
$$Q_l = \sum_{v=1}^{Z} |y_l(v)|^2 \quad (5)$$

$$U_l = \frac{Q_l}{\sum_{l=1}^{J} Q_l} \quad (6)$$

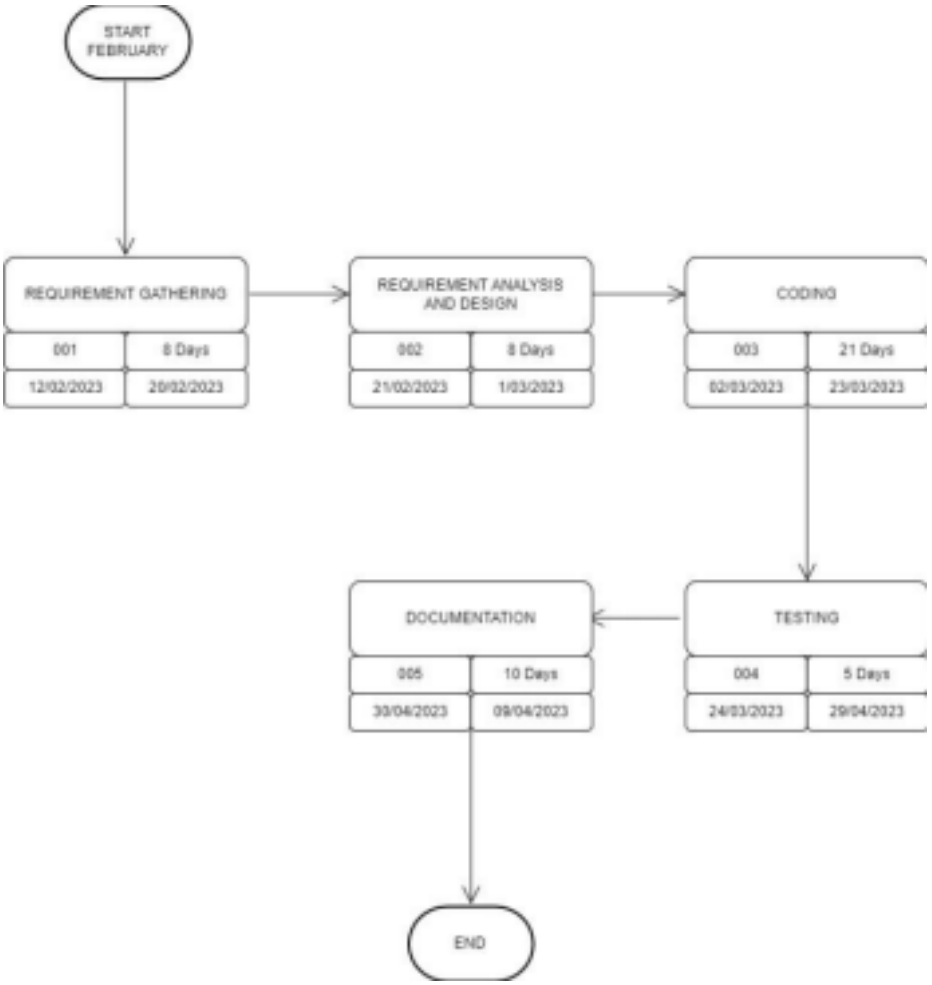
$$W_E = -\sum_{l=1}^{J} U_l \cdot \log(U_l) \quad (7)$$

\

UML Diagram


Tracks
-Tracks: arrayList<Track>
-TrackListName: String
-Mood: String
+GetAlbum()
+GetArtist()
+GetPlaylist(String mood)
USER
+takePhoto()
+moodSelect()
+approveMood(String Mood)
+controlMusic()
+editTag()
Device
+retrieveMood(): String
+generatePlaylist(String mood)
AccessPicture
+ClickPhoto()
+ScanImage()
+DetectFace()
+ExtractFeature()
MusicPlayerControls
+play()
+pause()
+shuffle()
+repeat()
+nextTrack()
+previousTrack()

Pert Chart


START
FEBRUARY
REQUIREMENT GATHERING
001
8 Days
12/02/2023
20/02/2023
REQUIREMENT ANALYSIS
AND DESIGN
002
8 Days
21/02/2023
1/03/2023
CODING
003
21 Days
02/03/2023
23/03/2023
TESTING
004
5 Days
24/03/2023
29/04/2023
DOCUMENTATION
005
10 Days
30/04/2023
09/04/2023
END

Results

[109]:
```
accuracy = (correct/total)*100
print(f'The accuracy for the proposed model is {accuracy}%')
```

```
The accuracy for the proposed model is 89.75%
```

```
tester = y_pred[0]
indices = []
for y in range(26):
    if tester[y]==1:
        indices.append(y+1)
emotion_ind = random.choice(indices)
```

```
emotion = None
for key,value in emotion_mapping.items():
    if value == emotion_ind:
        emotion = key
print(emotion)
```

```
Anticipation
```

```
playlist = emotion_playlist[emotion]
print(playlist)
```

```
https://open.spotify.com/playlist/0IusjjCCu02UMxOlrWLc1P?si=a4dcc9d2cf2c4762
```

Conclusion

In several disciplines, including psychology, marketing, and human-computer interaction, the development of emotion recognition systems has showed considerable promise. These systems are capable of correctly identifying emotions from facial expressions, voice, and physiological signs, which can offer important clues about a person's emotional state. There are still several issues that must be resolved, such as ensuring the precision and dependability of these systems in practical contexts, dealing with privacy issues, and resolving technological constraints.

Overall, emotion recognition systems have great potential benefits, and with more research and development, these systems may have a substantial influence on a variety of fields and applications.

References

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